SBI4U: Grade 12 University Biology

Metabolic Processes Assignment: Yeast Fermentation in Action

Overview & Purpose

In this unit, we have been exploring how cells harvest energy from organic molecules. While aerobic cellular respiration yields a large amount of ATP in the presence of oxygen, organisms like yeast (*Saccharomyces cerevisiae*) can undergo **anaerobic respiration (alcoholic fermentation)** when oxygen is scarce or rapidly consumed.

In this assignment, you will apply your knowledge of metabolic pathways, cellular respiration, enzymatic activity, and fermentation to a hands-on project. You will showcase the cellular processes occurring at a microscopic level by documenting a real-world application or controlled experiment. You will choose ONE of the options below. You may work alone or with a partner (or up to 3 for option A).

Important Guidelines for Safety:
Please consult with your parents/guardians before cooking in the kitchen and seek their approval.

Ensure proper food handling/kitchen safety (if cooking).

Important Note on Supplies:
If you do not have access to ingredients **do not worry! All necessary ingredients and materials can be provided for you for Option B (with teacher approval).** Please speak directly with your teacher or send an email at least three days before the prep date so we can make a plan.

Option A: The Culinary Fermentation Video Challenge (Bread or Yeast Recipe)

Task: Select a bread or yeast-leavened recipe (bread, pizza dough, cinnamon rolls, pretzels, etc).

Execution: Prepare the recipe at home while filming a video of yourself going through the steps:

1. **Preparation & Activation:** Proofing the yeast (waking up the dormant yeast in warm water/sugar).
2. **Mixing & Kneading:** Combining ingredients and developing gluten.
3. **Fermentation / Proofing Phase(s):** Observing and measuring the dough rising over time.
4. **Baking / Processing:** Documenting the thermal transformations in the oven.
5. **Sampling & Reviews:** Tasting the finished product yourself and collecting live, on-camera reviews and feedback from friends or family members.

Biochemical Voiceover/Commentary: Throughout the video, you must provide clear voiceover explanations (or text overlays/embedded scientific commentary) explaining the biology occurring at each step. You must be present in the videos (hands, face, etc).

Option B: Independent Experimental Inquiry

Task: Design and perform a controlled experiment testing a variable that affects the rate of yeast fermentation.

Possible Variables to Test:

- Type of substrate (e.g., sucrose vs. glucose vs. fructose vs. artificial sweeteners).
- Substrate concentration.
- Temperature of the environment/water bath.
- pH levels (e.g., adding lemon juice or baking soda).
- Presence of enzyme inhibitors or preservatives (e.g., salt concentration, spices).

Execution: Video document your experimental setup, procedure, quantitative data collection, and analysis. You will explain your hypotheses, controls, variables, and findings using metabolic terminology.

Biochemical Voiceover/Commentary: Throughout the video, you must provide clear voiceover explanations (or text overlays/embedded scientific commentary) explaining the biology occurring at each step. You must be present in the videos (hands, face, etc).

Option C: Student-Designed Yeast Product Pitch - must be pre-approved!

Task: Pitch and carry out a unique product or concept involving fermenting yeast outside of standard yeast breads (e.g., sourdough starters, fermented doughs from around the world). Same video requirements for Option A.

Mandatory Scientific Content Requirements (All Options)

Regardless of which option you choose, your video must explicitly incorporate and explain the following Grade 12 Biology concepts:

1. **Chemical Equations:**
    a. Provide the word equation for alcoholic fermentation.
    b. Compare the net ATP yield of alcoholic fermentation to full aerobic respiration.
2. **Metabolic Pathway Mechanics:**
    a. Glycolysis: Explain where it occurs in the yeast cell, substrate inputs and products.
    b. Fermentation Step: Explain all of the steps involved in the process.
3. **Biochemical Function in the Context:**
    a. ***For Culinary Options (A & C):*** How the dough expands; what happens to the ethanol during the baking process; why temperature control during yeast activation matters.
    b. ***For Experimental Option (B):*** How your specific variable impacts enzyme kinetics, metabolic rates, or membrane transport.
4. **Reflective / Analytical Component:**
    a. Explain why yeast undergo fermentation rather than aerobic respiration in specific conditions.

Video Submission Guidelines:

- ☐ **Length:** Edited to 7 - 10 minutes.
- ☐ **Format:** Video should be saved in your google drive with the sharable link added to the submission document. Ensure you have set the right sharing permission for your teacher.

Evaluation Rubric: Yeast Exploration Lab

Achievement Category	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)
Knowledge & Understanding **[10 marks]** ***(Understanding of metabolic concepts, biochemical pathways, and chemical equations)***	Demonstrates limited understanding of glycolysis and alcoholic fermentation pathways; chemical equations are incomplete or inaccurate.	Demonstrates some understanding of glycolytic and fermentation pathways; chemical equations are partially correct.	Demonstrates thorough understanding of glycolytic and fermentation pathways (NAD^+ regeneration, ATP yield, CO_2, ethanol); chemical equations are correct.	Demonstrates an insightful, thorough, and highly accurate understanding of metabolic pathways and biochemical details.
Thinking & Investigation **[10 marks]** ***(Execution of culinary/ experimental methodology, scientific reasoning, data gathering)***	Follows procedures with limited planning; provides minimal observation or analysis of the fermentation process; controls/ variables poorly addressed.	Follows procedures with moderate success; observations are basic; attempts to analyze fermentation mechanisms with partial accuracy.	Successfully plans and executes recipe/ experiment; collects clear qualitative/ quantitative data; analyzes the fermentation process accurately.	Demonstrates exceptional skill in experimental or culinary planning; collects rigorous data/ observations; provides insightful scientific analysis and problem solving.
Communication **[10 marks]** ***(Clarity, video production quality, sensory reviews, use of proper biological terminology)***	Video presentation is disorganized or unclear; biological terminology is rarely used or used incorrectly; reviews/sampling are absent or vague.	Video presentation is somewhat organized; terminology is used correctly occasionally; sensory reviews are basic.	Video presentation is well-structured, clear, and engaging; biological terminology is used correctly throughout; includes clear taste reviews/ feedback.	Video presentation is highly polished, creative, and professional; seamless integration of accurate biological terms; engaging sensory reviews/interviews.
Application **[10 marks]** ***(Connecting cellular biochemistry to real world applications, temperature/ environmental effects)***	Makes limited connections between microscopic yeast metabolism and macroscopic results (bread rising, gas production, baking effects).	Makes basic connections between microscopic metabolism and macroscopic properties of the product/ experiment.	Effectively connects biochemical processes (e.g., enzyme denaturation, CO_2 expansion, ethanol vaporization) to real world outcomes.	Flawlessly connects cellular-level metabolic mechanisms to real world culinary science, industrial applications, or biological phenomena.